ELI MKXI — Current State Snapshot (2026-06-19)

Contents

Authoritative, freshly-measured summary of the whole project after this session's work. Numbers are measured on ELI's real interpreter (`.venv/bin/python`), not the agent's system python. Deeper detail lives in the companion blueprints (`project_overview.md`, `architecture.md`, `orchestration_and_agents.md`, `grounding_and_evidence.md`, `capabilities_and_actions.md`, `complete_findings.md`).

Scale (measured 2026-06-19)

- **140,033 LOC** across **364** `.py` files under `eli/`.
- **208** manifest capabilities (`capability_manifest.json`; **183** routable): **174** in the executor `SUPPORTED_ACTIONS`, plugin-backed remainder.
- **12 main GUI tabs** (Chat, Proactive, Images, Quick Actions, Screen, Files, Labs, Coding, Tasks, Report Builder, Eli's World, Settings — Test & Review + Orchestration are Labs sub-tabs); **14** bus agents + the CodeAgent; **4** SQLite stores (user / agent / system_index / coding_memory).
- **166** test files.
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Tests & eval (measured on `.venv`, 2026-06-19)

- **6,924 tests collected**; full run **6,880 passed, 0 failed, 2 xfailed, 42 skipped** (GPU .venv).
- Composition: original unit/regression/integration (~2,600) + **`tests/claims/`** (~4,300 — examines the project vs its claims: every module compiles + core imports; every manifest capability well-formed + flags match the live executor; every SUPPORTED_ACTION handled; every documented action reachable; activation phrases; blueprint refs resolve; structural/behavioural claims; **symbol inventory** = every public function/class/method is a real introspectable callable) + the meta-tests for the doc pipeline, evidence planner, and test generator + generated tests.
- **Eval harness:** 93 cases in `cases.yaml` = **60 model-free router (auto-run under pytest, green)**
  - **engine cases (model-backed)**. The engine board + full report now refresh **nightly** (scheduled `eval` task), as does ELI-assisted test generation (`testgen`).
- **2 xfail = documented, intentional:** `MOUSE_CONTROL` bare "left click" → GAZE_CLICK by design; one grounding-mode case.

GPU / inference (corrected this session)

ELI runs on the **GPU**. `.venv/bin/python` → `llama_cpp.llama_supports_gpu_offload()` = **True** (llama-cpp-python 0.3.23, CUDA; driver 595.71.05, CUDA 13.2, nvcc 12.8). Model loads with `ggml_cuda_init: found 1 CUDA device`, layers offloaded to the 2060 Super; a short generation ≈ **1.2 s**. (An earlier "CPU-only" finding was a sandbox error — the agent's bare `python` is the *system* interpreter with a CPU wheel, not the venv ELI uses. No rebuild needed.)

What this session added / fixed (all on `main`, green)

- **Grounded generation:** evidence-planner (plan→gather→consume across code/web/ memory/runtime agents) + multi-stage `report_pipeline` (outline→sections→review) for documents/scripts/projects, with confidence-driven deeper-tier re-gather.
- **Grounded perception/introspection:** "what's on screen" → real vision glance; identity/awareness/cognition queries gather-then-summarise (no data dumps, no weights-only answers).
- **Autonomy/self-awareness tick** wired into the proactive daemon (governed, 30-min: code-monitor + self-model refresh + goal/scheduler proposals).
- **Self-correctness loop:** auto test report on every pytest run (`artifacts/test_report.md`); `RUN_TESTS` + `GENERATE_TESTS` actions ELI can run and summarise; the report is an evidence channel for upgrade proposals; nightly `eval` + `testgen` scheduled (durable + recurring).
- **GUI:** Report Builder promoted to a main tab; Files-tab document converter (pdf/docx/doc/odt/rtf/html/md/tex/e + lualatex).
- **Fixes:** habit 00:00 offer; EXAMINE_CODE↔GUI_RUNTIME_AUDIT route collision; grounding quick-mode hedge regression; 4 router-gap fixes; venv test-collection error; scheduled-store test isolation. All prior known failures resolved.

Honest weaknesses (unchanged; from `complete_findings.md`)

- **God-files:** `executor_enhanced.py`, `engine.py` (~13–14k each) + the GUI (~11k).
- **~2,565 swallowed `except Exception`** — failures are absorbed, not surfaced.
- **~15 load-bearing `globals()`/install-time wrappers** (netguard, adaptive GGUF loader, middleware table) — architecture, not test-maskers, but a fold-in target.
- **Duplication:** two image engines; overlapping `runtime/*_response/*_surface`.
- **Repo hygiene:** root junk (`...`, one-off `patch_*.py`, `verify_*.sh`, zips).
- **The real ceiling is the local model** — the body/memory/grounding/awareness are frontier; the *mind* is whatever local GGUF is loaded (model-agnostic → a swap).

Verdict

The architecture is genuinely frontier for a local, model-agnostic, self-honest personal AI — it tests itself, evals itself nightly, writes its own tests, grounds its generation in real evidence, resolves intent with the model against its own catalogue, hears its wake word over music with a model it trained itself, and reads the user's vocal tone into its responses — all on the GPU, all local. The open work is engineering debt (god-files, swallowed exceptions, the verbatim/routing logic duplicated across GUI/engine/ router) and the model ceiling — both known, both phased, neither blocking.